Short report on the tree-ring and phenology data for the Tree Spotters

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1 How we collected the cores, processed, scanned and measured them

1.1 Tree coring at the Arboretum

At the end of April 2025, Lizzie, my PhD colleague, Mao and I went to the Arboretum to core the trees the Tree Spotters have been monitoring for the last nine years (Figure 1). Even though coring has very little effect on the tree health, we couldn't core some trees, which were mostly American beech. Fifty trees were accepted in the proposal!

Figure 1: Mao extracting a core from a White oak

During the three days it took to coring the trees, we had the chance to present our work and project objectives to the Tree Spotters (Figure 2). This presentation was particularly great because the Trespotters shared their experience and anecdotes with their trees that they have been monitoring for years (Figure 3)!

Figure 2: Mao and Christophe coring a White oak during a short presentation with the Tree Spotters

Figure 3: Shanon H. looking at an early emerging bud

1.2 Core mounting and sanding

Once all the cores were collected, they were left to dry at the Weld Hill building. In July, we went to pick them up so we could proceed with mounting them. We mount the tree cores on a small wooden structure to straighten the core and also to avoid breaking it. Then we sanded the cores from a rough to a progressively finer grit, which exposes the rings nicely (Figure 4)

1.3 Core scanning and ring measurements

Once all the cores were properly sanded, we scanned them at a very high resolution using our lab-made tree-ring imaging machine (TIM), which our undergraduate students created. This device has a microscopic lens attached to it that allows us to zoom in a lot onto the core to measure the ring width (Figure 5). With these cores scanned, we were able to measure each ring width for each core on a computer (Figure 6).

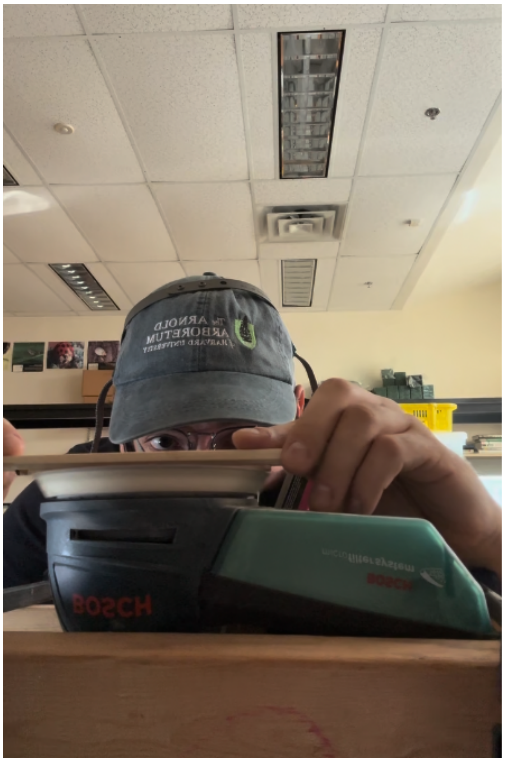


Figure 4: Christophe sanding a core.

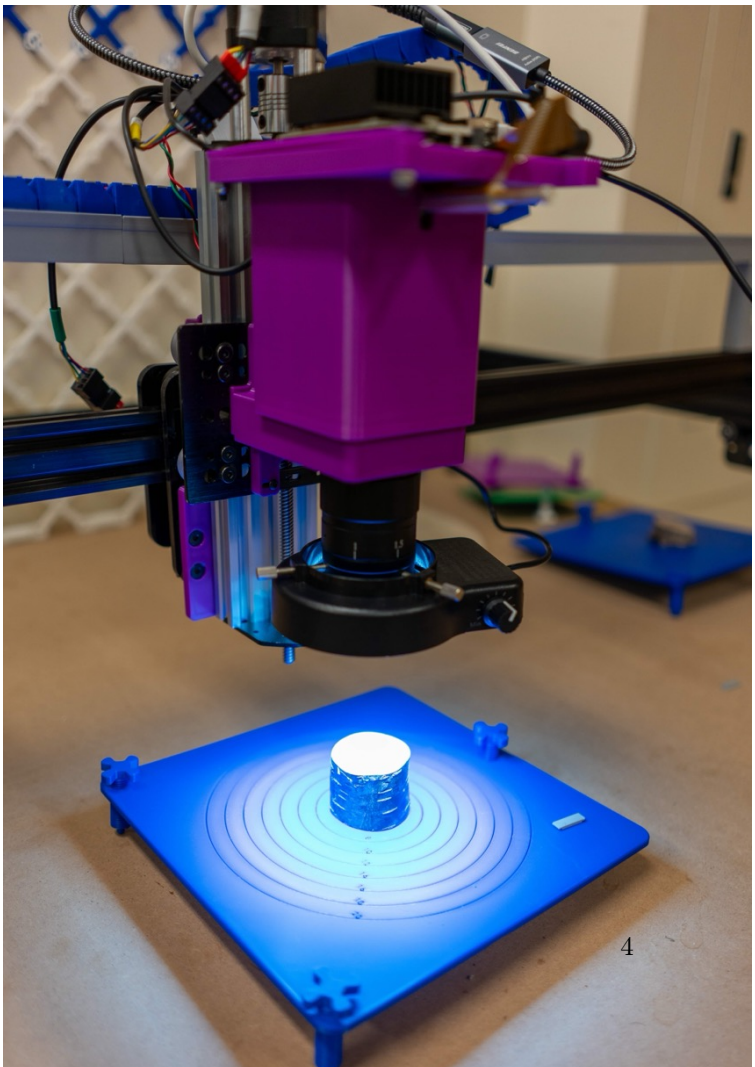


Figure 6: An example of what the scanned cores look like when I measured them. This is CAOV 12907*D.

1.4 Crossdating

Dendrochronology is the study of tree rings. In that field, statistical crossdating is standard and consists of dating the tree rings to the exact calendar year by comparing a chronology to other chronologies. Often, marker years are used because they represent a particularly narrow ring, which can be the result of a severe drought or pest outbreak. However, statistical crossdating requires a sufficient number of trees of the same species, and a chronology at least 50 years old. These criteria were often not met with our cores. Thus, after discussing with Neil Pederson about this, we decided it would be hard to perform statistical cross-dating on those cores. We ended up only visually cross-dating, which consists of visually matching the years across the cores of the same species (Figure 7).

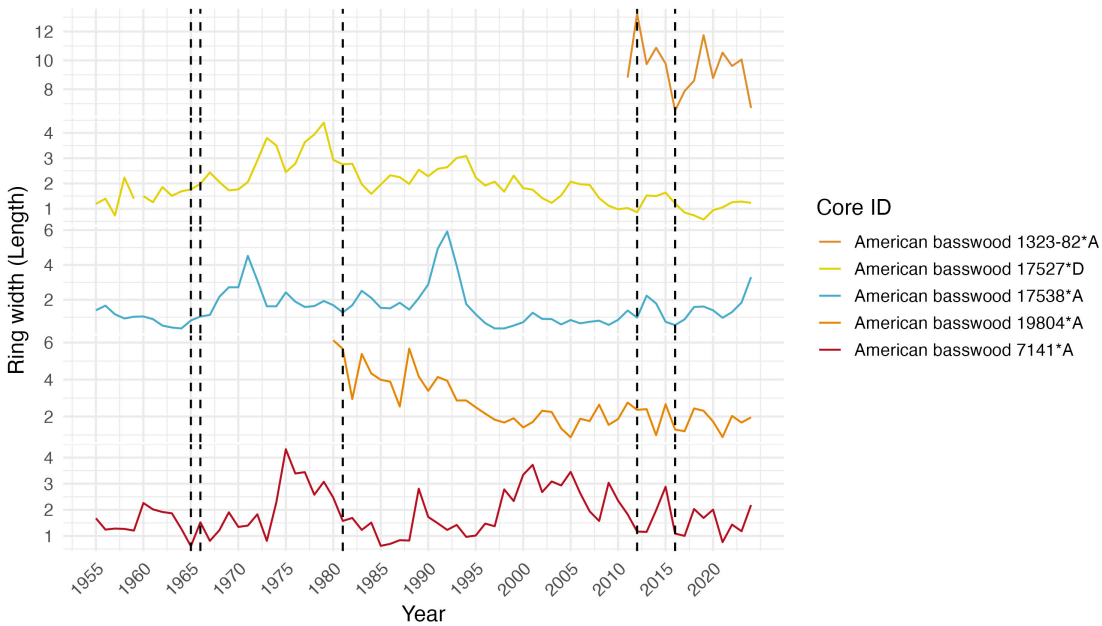


Figure 7: Example of what visual crossdating looks like. This is all the American basswood (*Tilia americana* chronologies that we had. The dashed lines represent marker years, for which our trees did not appear to respond.

2 Phenology and tree-ring results

With climate change, trees are shifting their phenology earlier in the spring and later in the autumn, thus extending the growing season. This is widely expected to increase tree growth, with important effects on forest carbon sequestration dynamics. Multiple recent studies, however, have failed to find this relationship. Therefore, we will address this decoupling by using the data collected by the Tree Spotters, which will allow us to investigate how the timing of phenological events affects growth across years for mature trees. We will do this by relating these observations to tree rings. We hope our results will provide a better understanding

of the recently observed decoupling between growth increment and longer growing seasons.

2.1 Phenology plots

We plotted the two phenophases we used to calculate the growing season conditions: leafout (Figure 8) and leaf colouring (Figure 9).

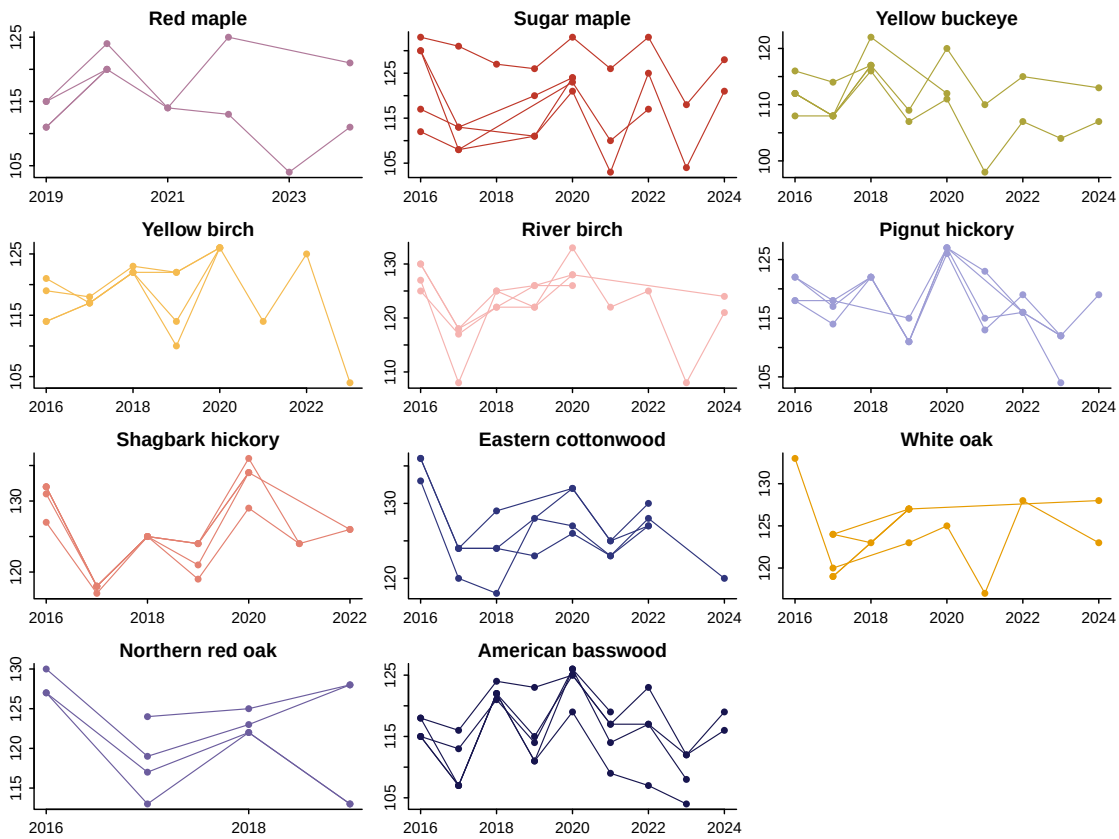


Figure 8: All the leafout dates recorded over the years, for each individual tree. The vertical axis represents the day of the year of that observation.

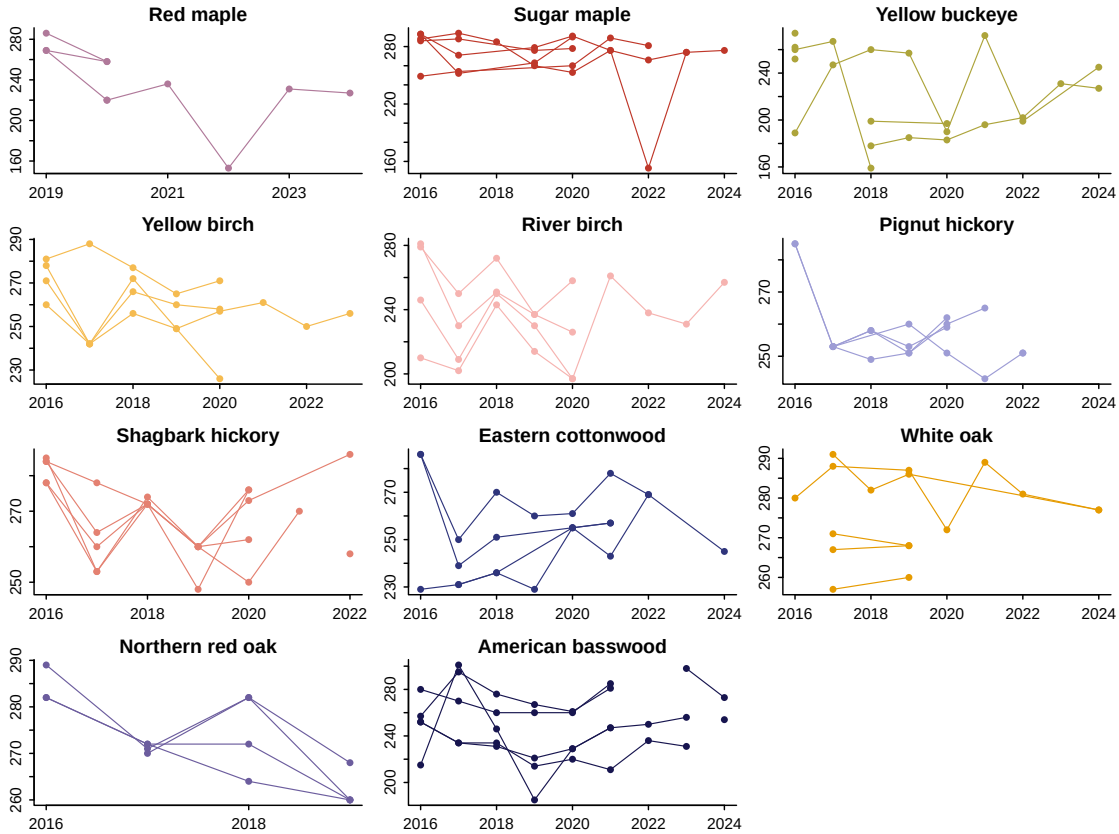


Figure 9: All the leaf colouring dates recorded over the years, for each individual tree. The vertical axis represents the day of the year of that observation.

2.2 Tree ring chronologies

We present the tree-ring width as a function of year, only for the years the Tree Spotters collected phenological data for (Figure 10).

Figure 10: The ring width measurements as a function of each year, for each individual tree, and the years we have phenological observations for. The vertical axis represents the ring width of that individual.

2.3 Preliminary results

Our results indicate that while some species, like maples (*Acer saccharum* and *Acer rubrum*), River birch (*Betula nigra*) and White oak (*Quercus alba*) appear to grow more with increasing growing degree days which is a metric of heat accumulation that sums the mean daily temperature above some threshold (here: $5°C$). Others, like Yellow buckeye (*Aesculus flava*), Yellow birch (*Betula alleghaniensis*), hickories (*Carya ovata* and *Carya glabra*) and American basswood (*Tilia americana*) grew less. The growth of the other species was not affected by growing degree days (Figure 11).

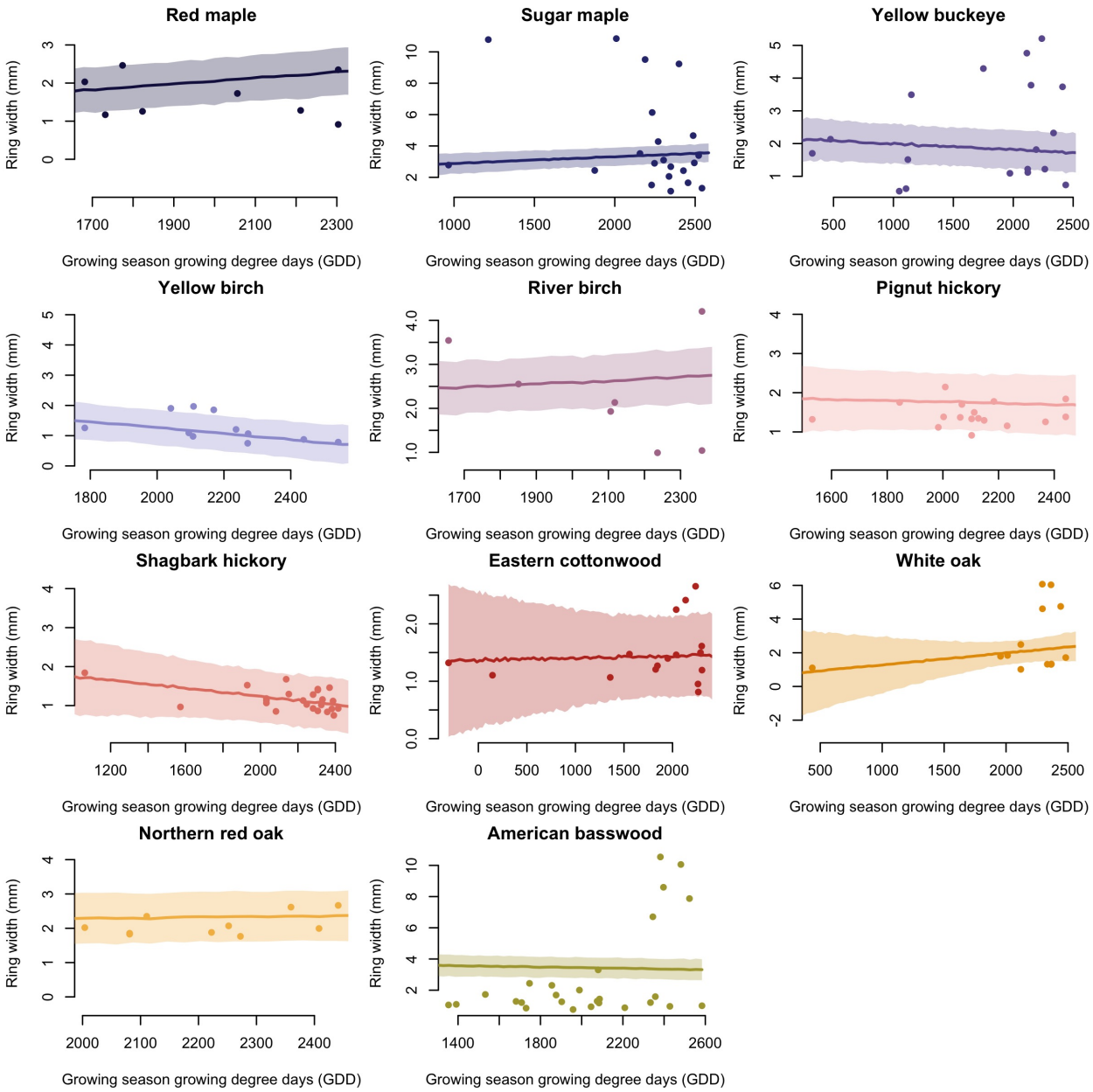


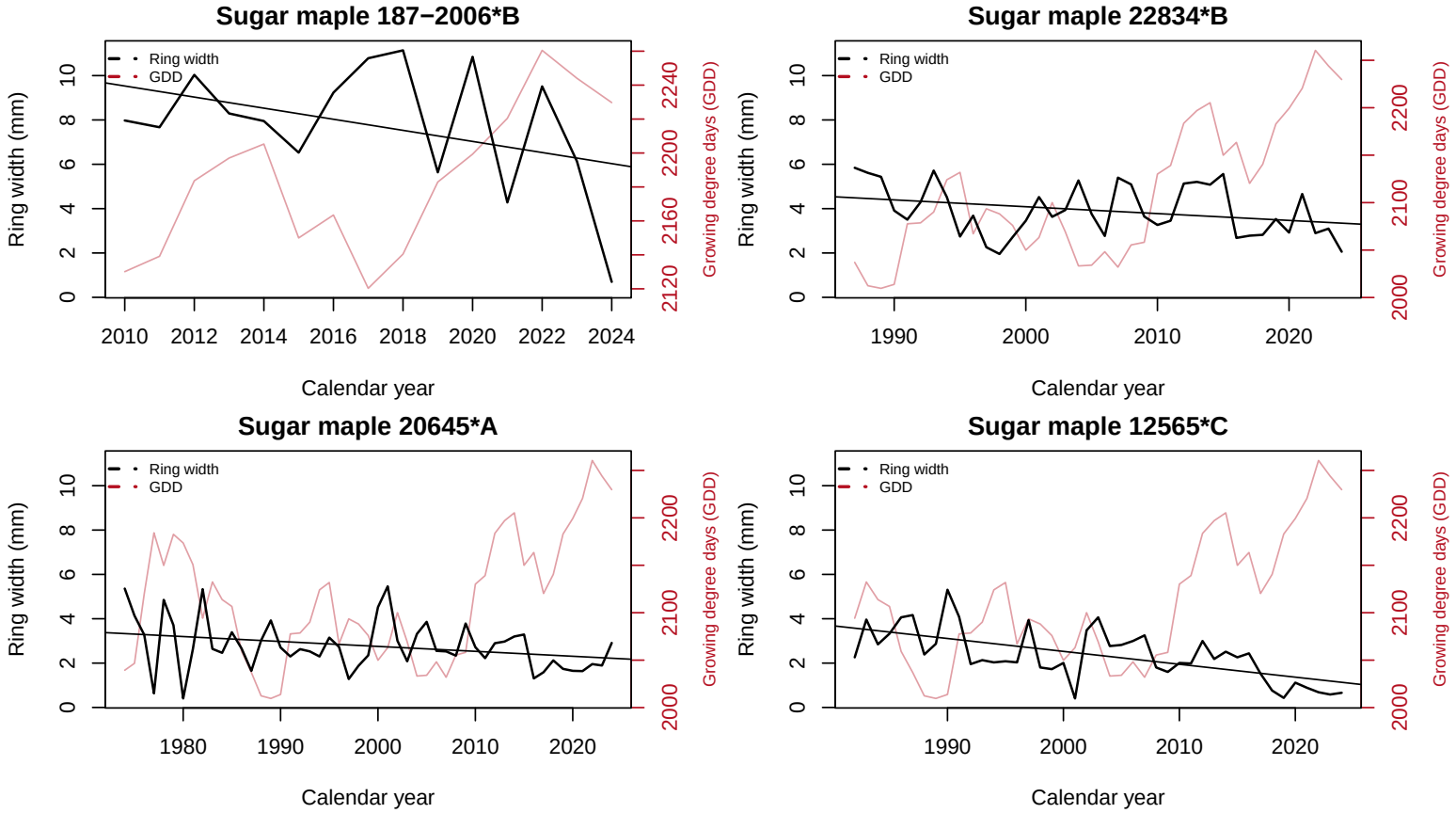
Figure 11: Model estimates for each species. We ran a model to estimate the ring width as a function of growing degree days (a metric of heat accumulation that sums the mean daily temperature above some threshold (here: $5°C$)) for the growing season defined between leaf colour date and leafout. The very low growing degree day values for Red maple (15350*A, 2022), Sugar maple (22834*B, 2022) and Yellow buckeye (12651*D, 2018) are the result of a very early leaf colouring observation in early June.

3 Supplemental material

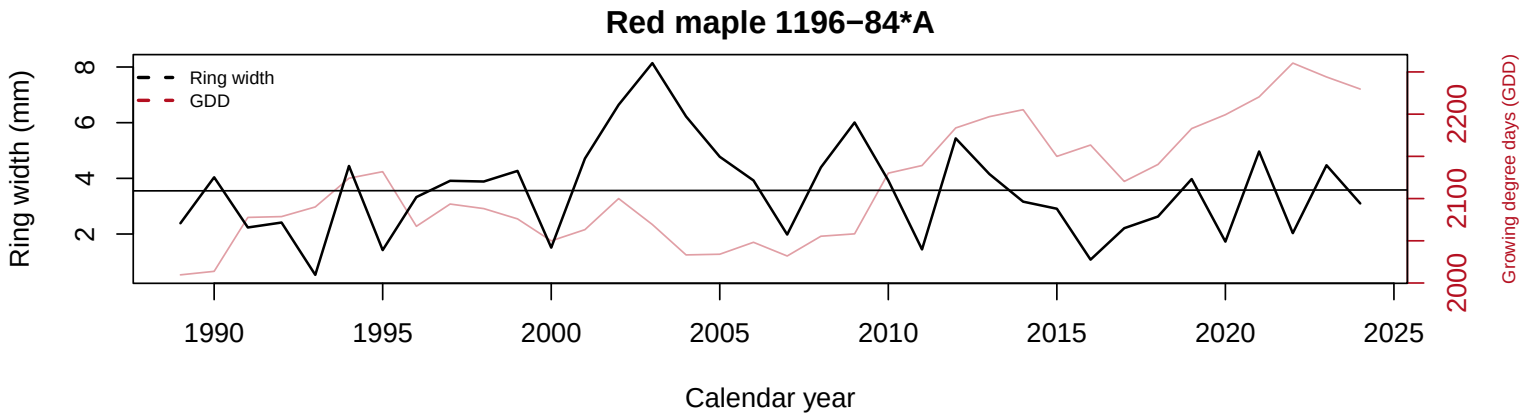
3.1 All the tree chronologies with a five-year averaged growing degree days

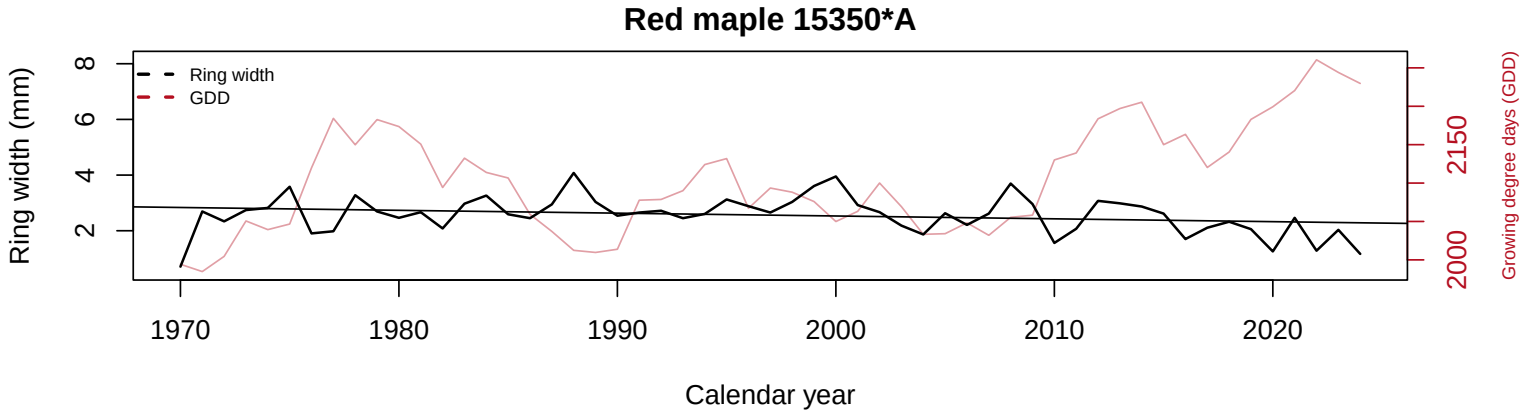
We present the whole ring width chronologies for each tree along with a five-year moving average of growing degree days.

Sugar maple

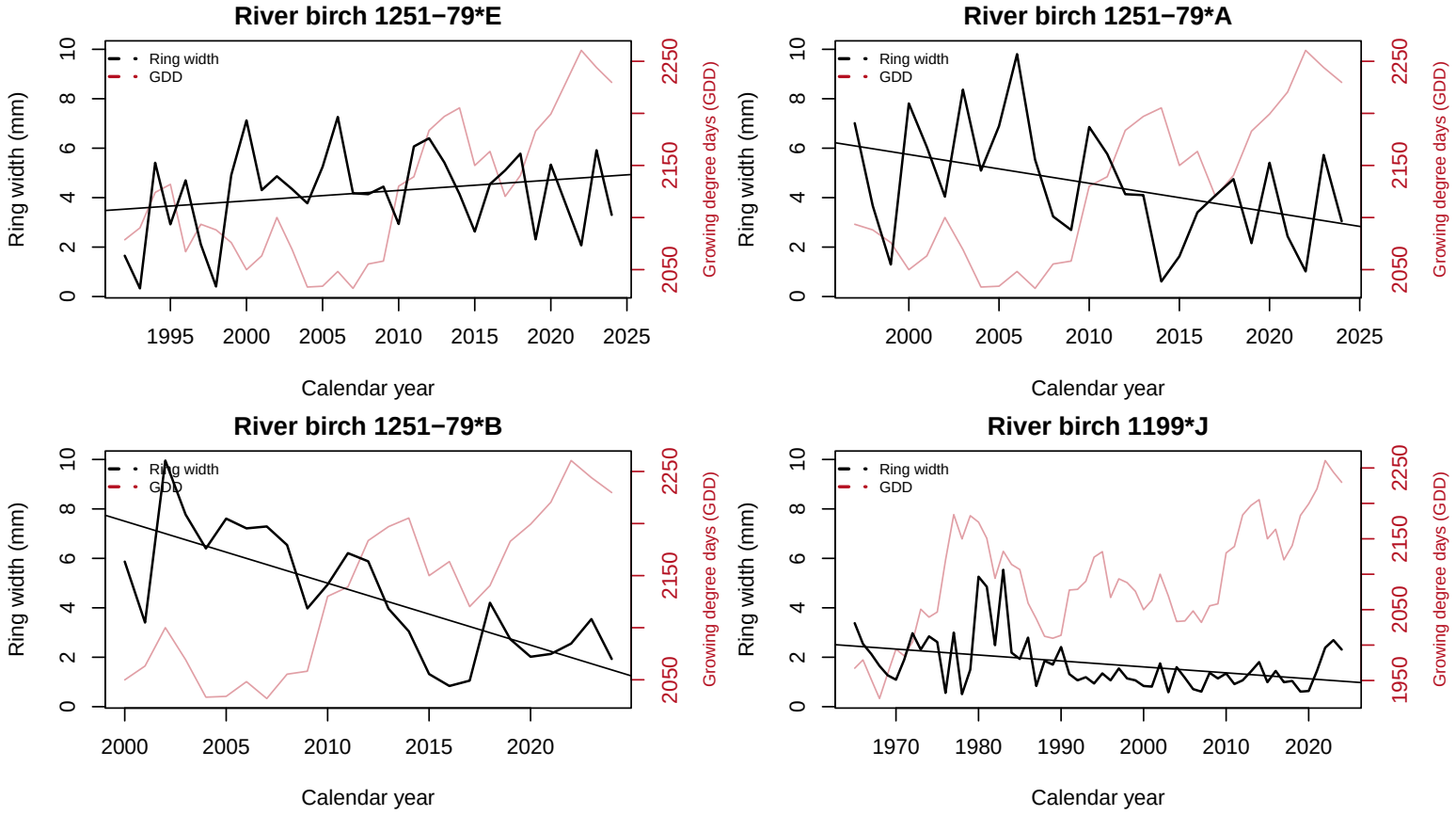

Sugar maple 187−2006*B
Ring width
GDD
Ring width (mm)
Growing degree days (GDD)
Calendar year
Sugar maple 22834*B
Ring width
GDD
Ring width (mm)
Growing degree days (GDD)
Calendar year
Sugar maple 20645*A
Ring width
GDD
Ring width (mm)
Growing degree days (GDD)
Calendar year
Sugar maple 12565*C
Ring width
GDD
Ring width (mm)
Growing degree days (GDD)
Calendar year

Red maple

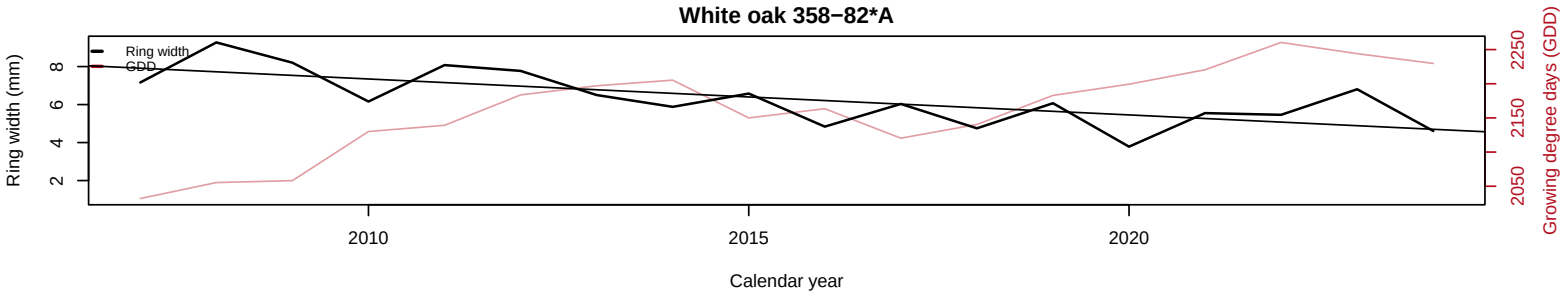

Red maple 1196−84*A
Ring width
GDD
Ring width (mm)
Growing degree days (GDD)
Calendar year
1990
1995
2000
2005
2010
2015
2020
2025
2000
2100
2200
2
4
6
8

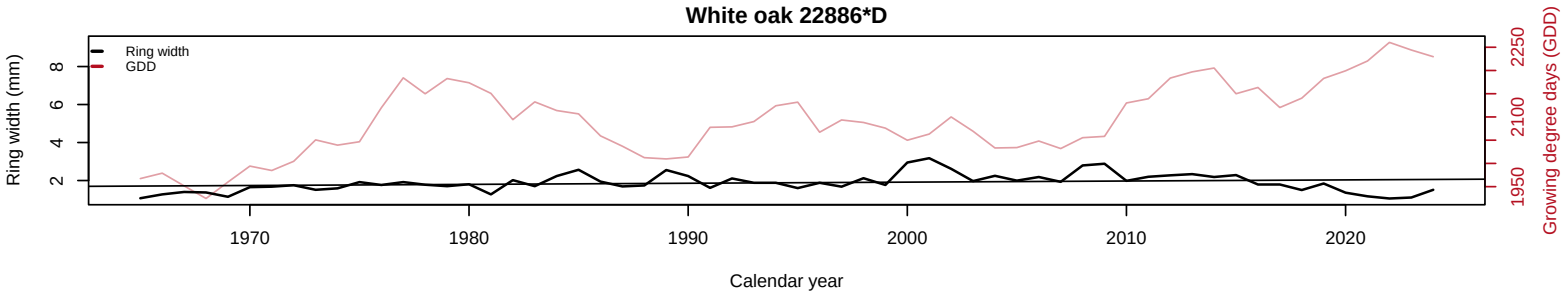

Red maple 15350*A
Ring width
GDD
Ring width (mm)
Growing degree days (GDD)
Calendar year
1970
1980
1990
2000
2010
2020
2000
2150
2
4
6
8

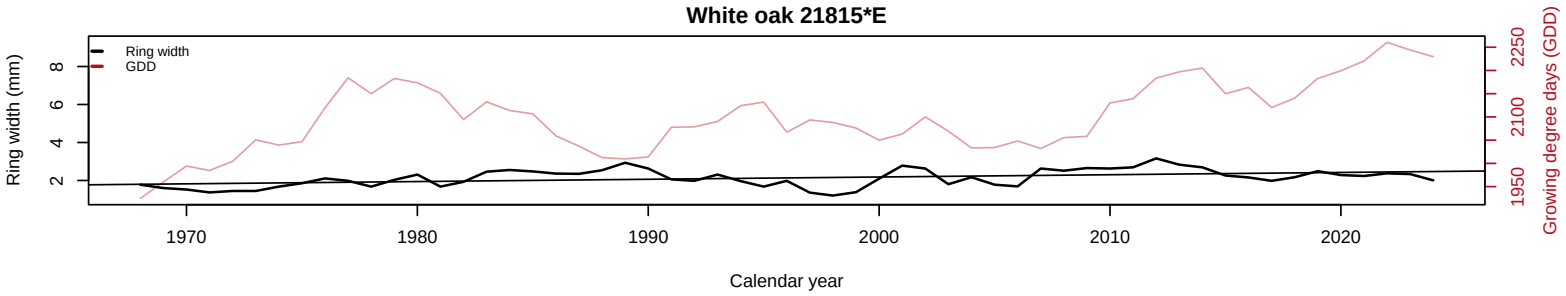
River birch


River birch 1251−79*E
Ring width
GDD
Ring width (mm)
Growing degree days (GDD)
Calendar year
River birch 1251−79*A
River birch 1251−79*B
River birch 1199*J

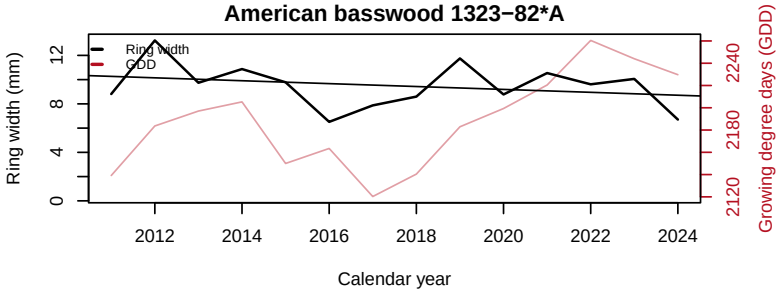
White oak

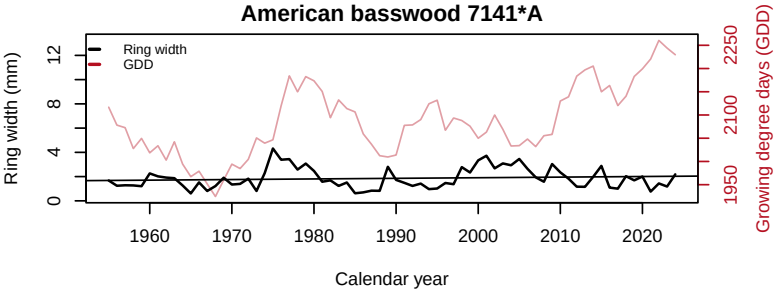

White oak 358−82*A
Ring width
GDD
Ring width (mm)
Growing degree days (GDD)
Calendar year

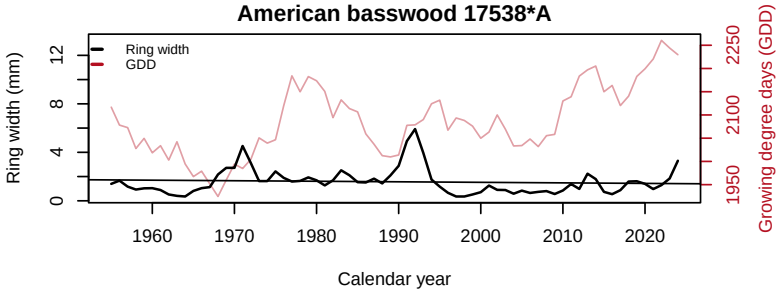

White oak 22886*D
Ring width
GDD
Ring width (mm)
Growing degree days (GDD)
Calendar year

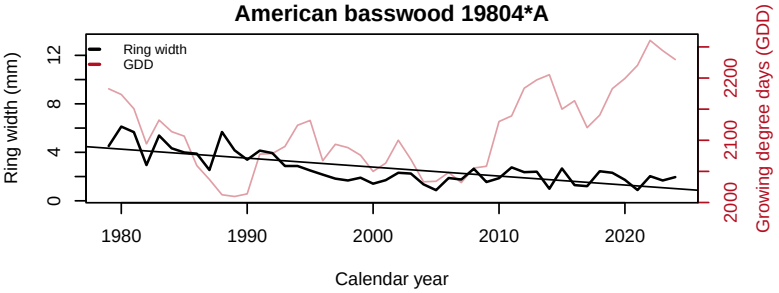

White oak 21815*E
Ring width
GDD
Ring width (mm)
Growing degree days (GDD)
Calendar year

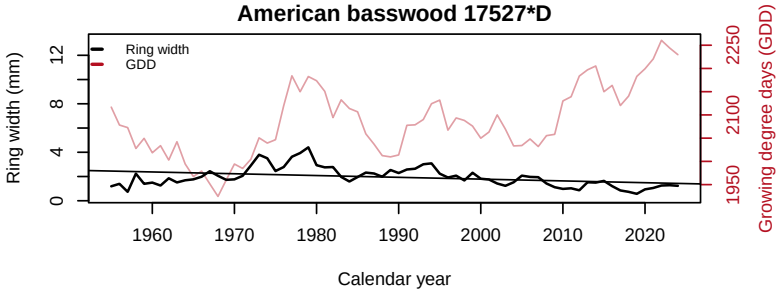
American basswood



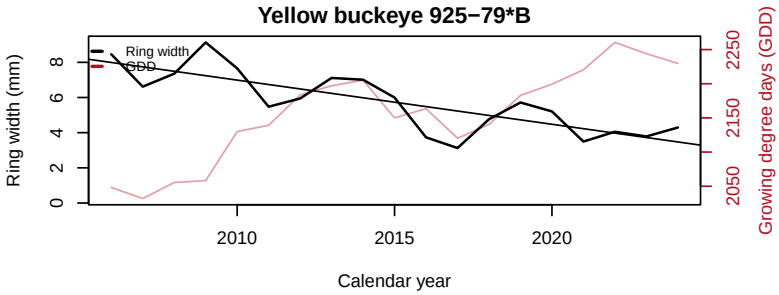


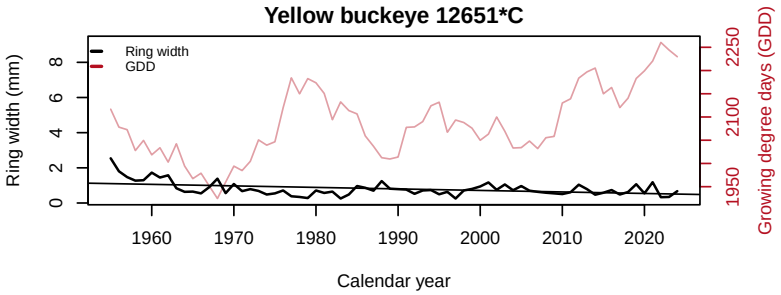


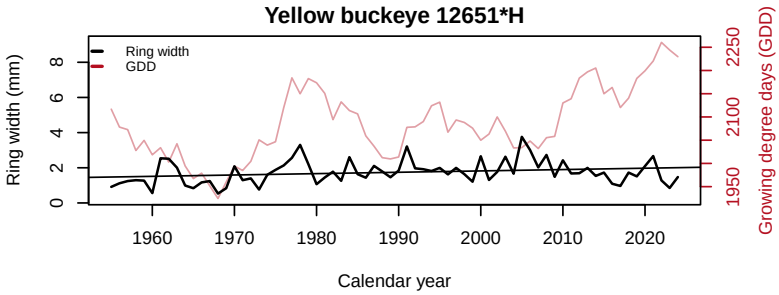


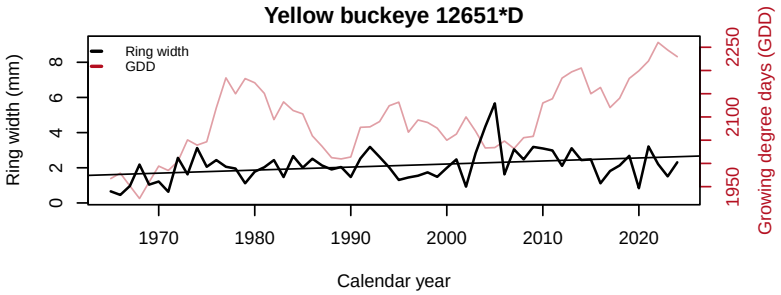


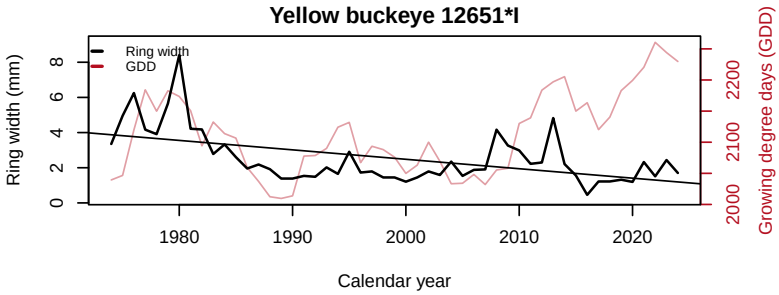
Yellow buckeye



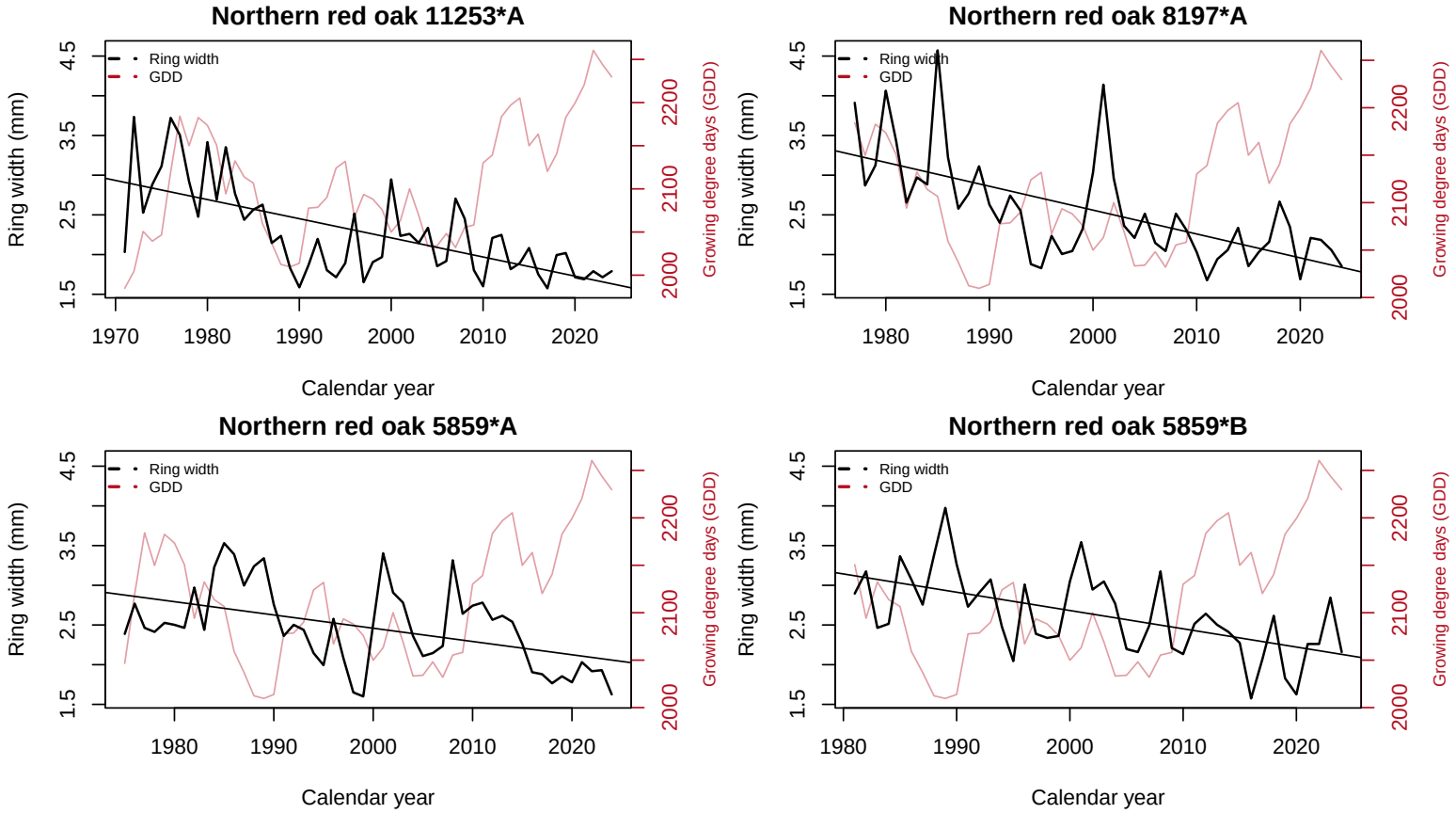




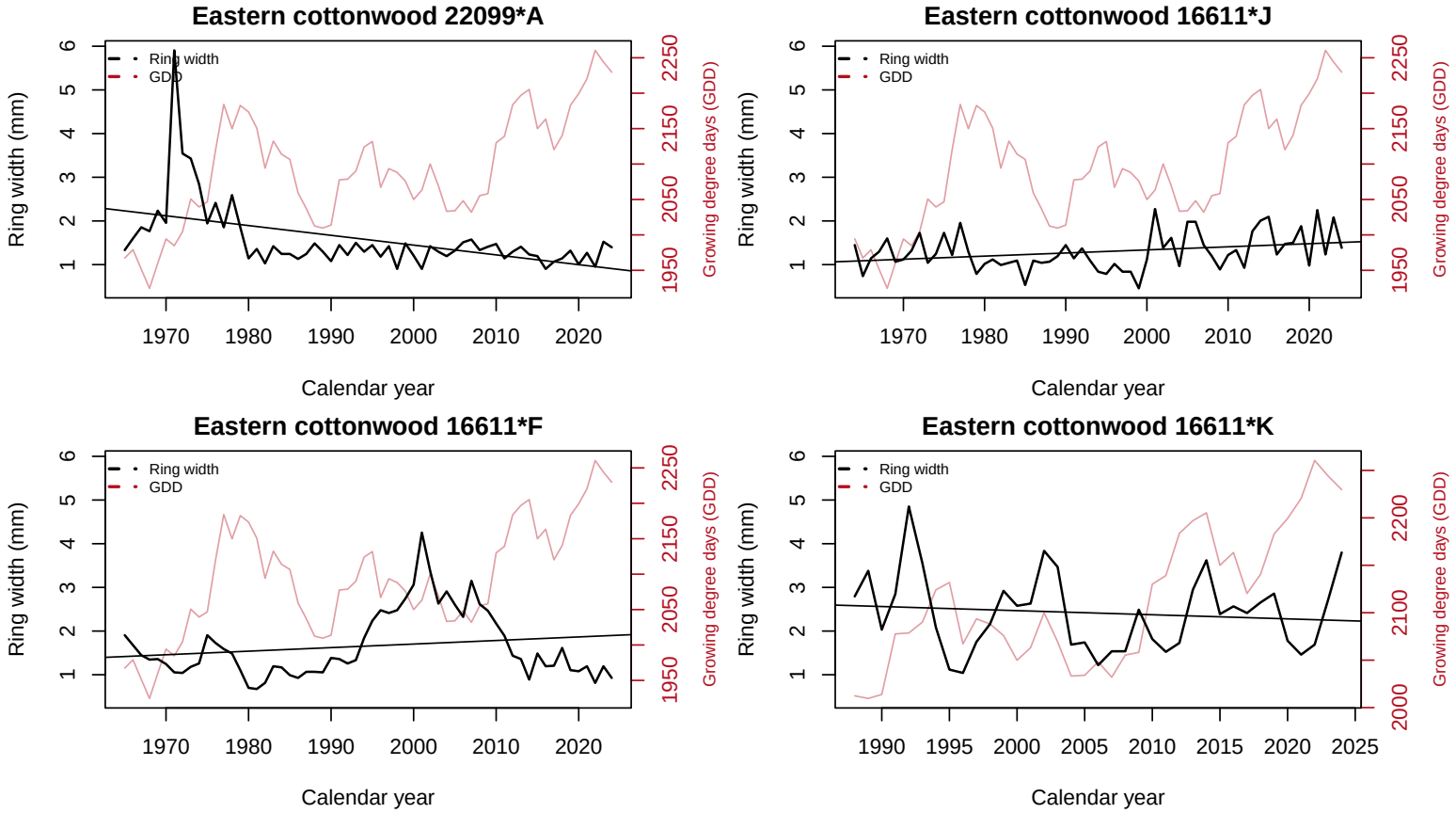




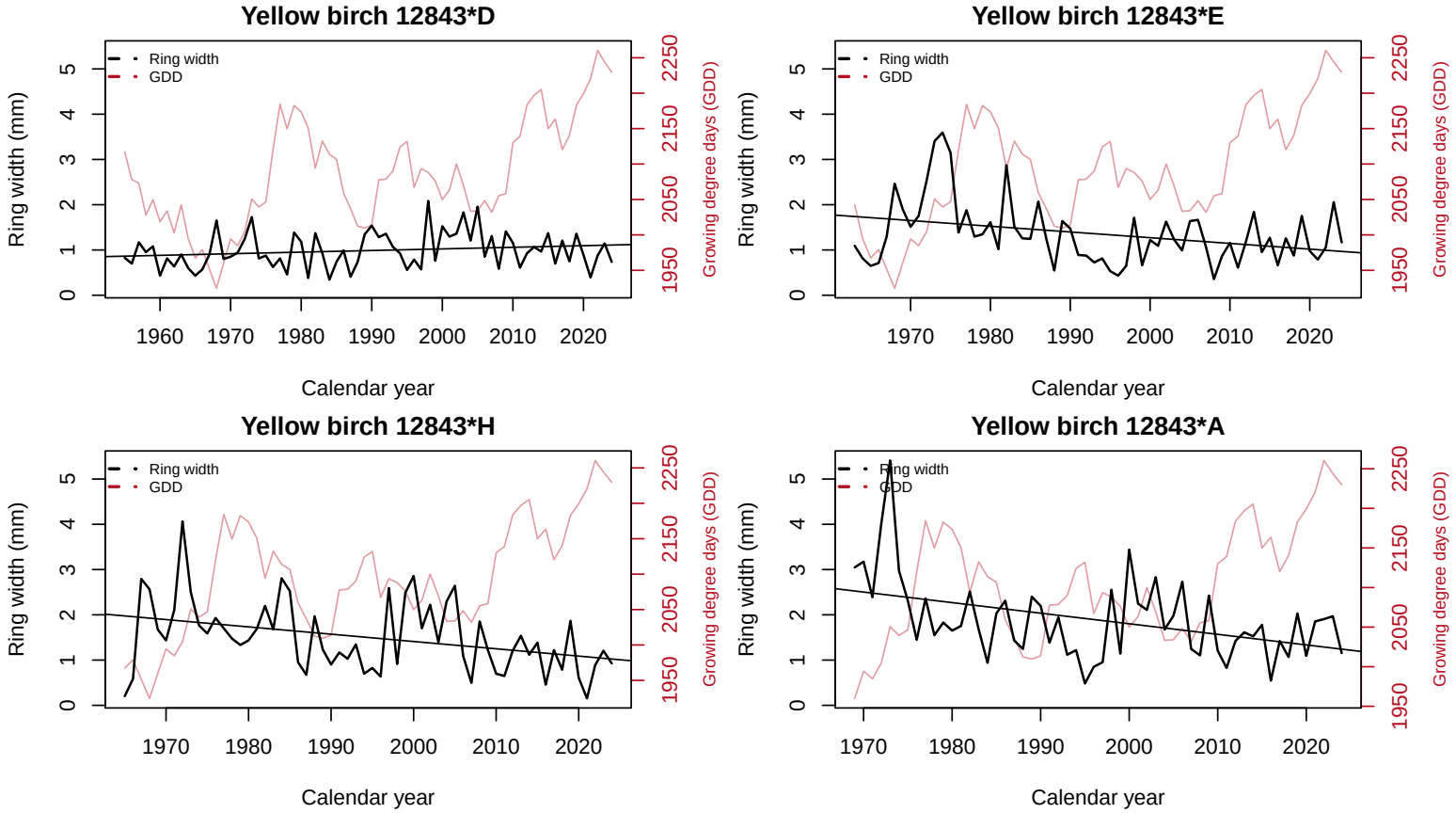
Northern red oak



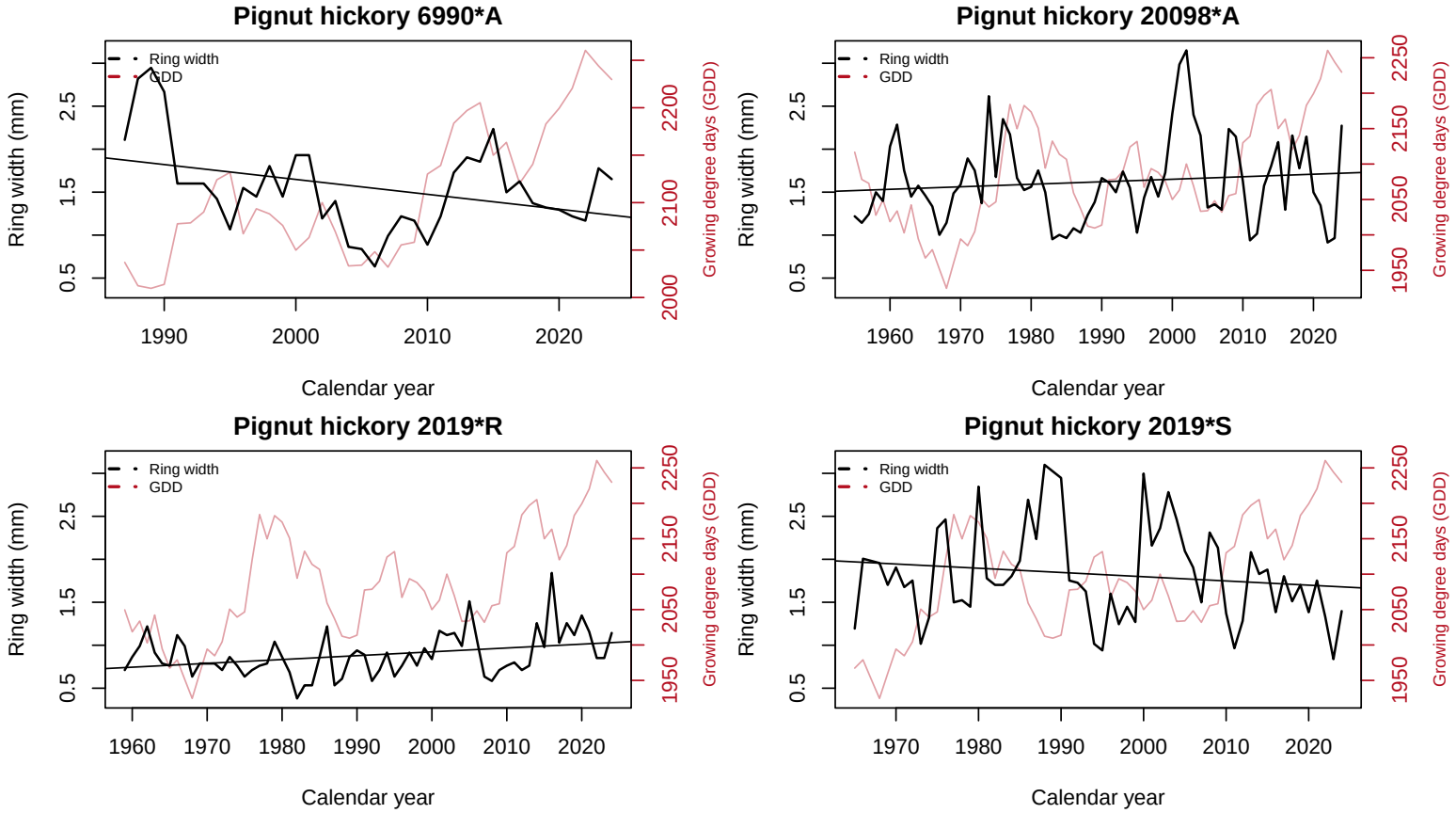
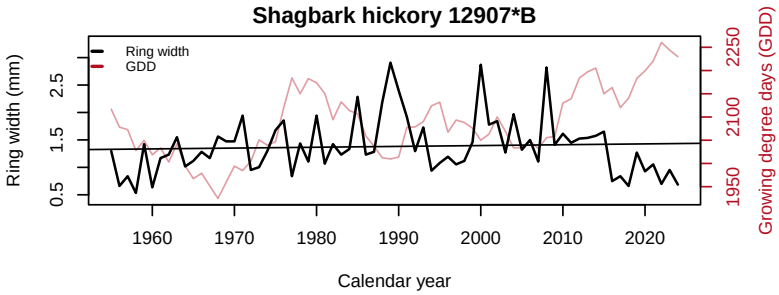
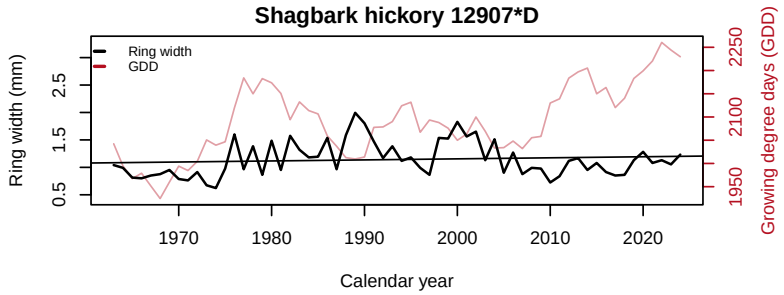
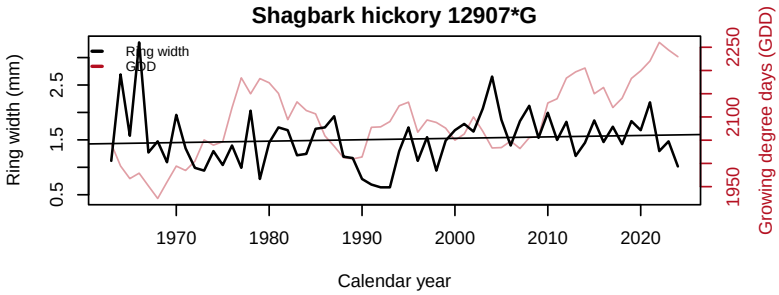
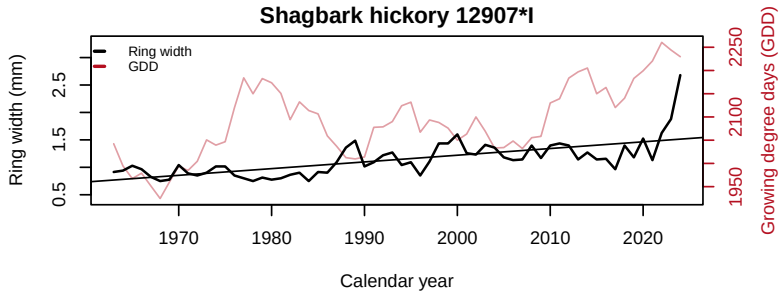
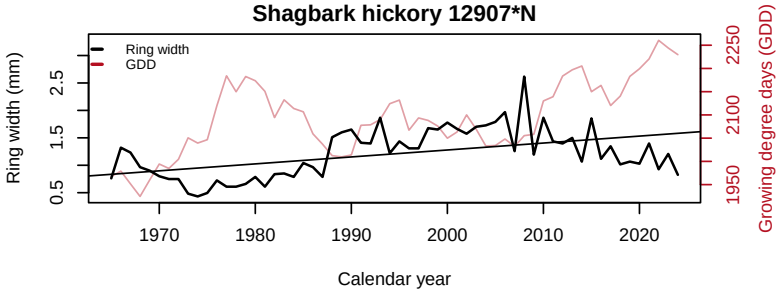
Eastern cottonwood



Yellow birch



Pignut hickory



Shagbark hickory